

Documentation WMA data input REMIX

This document contains the documentation for the WMA (Water Mixing Approach) data input in the code **REMIX**. This input consists of two files that complement the chemical data input files. These files define the time discretisation and the precomputed mixing ratios, so the code does not need to compute the mixing ratios internally. Both files must contain the **same root** in their names, and they must live in the **same directory** as the chemical input files.

We describe the input files with the same example of two waters that mix: an inflow water in equilibrium with atmospheric O_2 and CO_2 and a resident water in equilibrium with calcite. This example can be found in the directory `"..\examples\calcite_eq"`.

Time discretisation file: "_WMA_discr.dat"

This file contains the information related to the **time discretisation for the WMA method**. It defines the time weighting factors for reactions, the time step, and the number of time steps. It is read by the **Time Discretisation** class.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Line 3: Section name:** 'TIME DISCRETISATION'
 - **Line 3a:** *theta_r* (real)
 - *theta_r*: Time weighting factor for reactions (0: Euler Explicit, 1: Euler fully Implicit, 0.5: Crank-Nicolson)
 - **Line 3b:** *dt* (real)
 - *dt*: Time step (days)
 - **Line 3c:** *num_time_steps* (integer)
 - *num_time_steps*: Number of time steps
- **Line 4:** Section separation
- **Line 5:** end of file

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
'-----'
'TIME DISCRETISATION'
5d-1                ! Time weighting factor for reactions
1d-2                ! Time step (days)
100                 ! Number of time steps
'-----'
'end'
```

Mixing ratios file: "_WMA_lambdas.dat"

This file contains the **mixing ratios and mixing waters indices** used in the WMA method for concentrations. It defines how waters are mixed at each target of the domain. It is read by the **Transient Transport** class.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Line 3: Section name:** 'MIXING RATIOS'
 - **Lines 3a:** One line per target. Each line starts with the number of mixing waters (integer), followed by the mixing ratios (real values in scientific notation). The mixing ratios for concentrations include the target itself. A line with 0 marks the end of the section.
- **Line 4:** Section separation
- **Line 5: Section name:** 'MIXING WATERS'
 - **Lines 5a:** One line per target. Each line starts with the target index (integer), followed by the indices of the mixing waters used for concentrations (integers).
- **Line 6:** Section separation
- **Line 7:** end of file

```

'TITLE OF THE PROBLEM: Calcite in equilibrium'
'-----'
'MIXING RATIOS'
12 3.3581882094E-02 8.3954705235E-01 ... 1.0957134334E-08 0.0000000000E+00
12 5.3883119697E-03 1.3470779924E-01 ... 7.4265021594E-08 0.0000000000E+00
...
12 2.6675253721E-09 6.6688134302E-08 ... 8.3954705235E-01 0.0000000000E+00
0
'-----'
'MIXING WATERS'
1 11 1 2 3 4 5 6 7 8 9 10 12
2 11 1 2 3 4 5 6 7 8 9 10 12
...
10 11 1 2 3 4 5 6 7 8 9 10 12
'-----'
'end'

```